

A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 21

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- The number in brackets after each marking point is the mark it earns; the note above the points says how many are needed.
- Where a question asks for a fixed number of points (eg *two reasons*), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	<i>1 mark for each point.</i> • Igneous intrusions (batholiths) of magma cool within the crust, heating groundwater to produce superheated, pressurised water (1) • The hot water dissolves minerals from the surrounding rock and the mineral-rich solutions travel along fissures / cracks away from the batholith (1) • As the solutions cool the dissolved minerals crystallise in order of solubility (least soluble first) - fractional crystallisation (1) • So different metals are separated and concentrated in veins, eg tin, copper, lead, silver, gold, arsenic (SW England batholith) (1)	4
01.2	<i>1 mark for each point.</i> • Dissolved iron compounds in ancient seas were oxidised by the oxygen released by early photosynthetic organisms (2.5-1.8 billion years ago) (1) • Iron oxides are insoluble so they were deposited as sediments (haematite, magnetite) (1)	2
01.3	<i>1 mark for each point.</i> • Flowing water carries solid particles; the ability to carry them depends on the water velocity and the density of the solids (1) • As velocity falls, dense minerals (gold, tin ore) are deposited first while lighter material is carried further, concentrating the heavy minerals in particular places (1)	2
01.4	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Limestone / chalk from the shells of marine organisms (1) • Coal from terrestrial vegetation (1) • Crude oil and natural gas from marine organisms (plankton) (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 132-134; AQA spec 3.2.3.2.	

Question	Marking guidance	Marks
02.1	<i>1 mark for each point.</i> • Affluent people buy more consumer goods / trivia / disposable items and throw items away rather than repairing them (1) • Built-in obsolescence: products designed to have short lives / wear out / go out of fashion so customers upgrade (1) • Disposable products (razors, lighters, pens) that must be replaced (1) • Over-packaging to make products attractive; also increases wastes from manufacture and mineral extraction (1)	4
02.2	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Properties of the waste: toxicity, flammability, degradability, radioactivity (1) • Population density (collection costs) (1) • Land availability for landfill (1) • Availability of recycling technology / markets (1) • Legislation / regulatory framework (landfill tax) (1) • Public willingness to recycle / environmental awareness (1) • Processing costs / household income (1)	3
02.3	<i>1 mark for each point.</i> • Advantage: aerobic decomposition produces little methane; nutrients / organic matter returned to soil (circular economy); reduces landfill volume (1) • Disadvantage: needs separation / collection of clean organic waste; odour / pests / pathogens if poorly managed; low-value product; not suitable for mixed waste (1)	2
02.4	<i>1 mark.</i> • Many different materials are mixed together / contaminated so identification and separation are costly and the quality of recovered material is reduced (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 279-282; AQA spec 3.4.3.2.12.	
03.1	<i>A correct final answer (with unit where asked) scores full marks. If the answer is wrong, award 1 mark for each correct step of working shown.</i> • Conventional: $40 / 100 = 40\%$ (1) • CHP: $(35 + 45) / 100 = 80\%$ (1)	2
03.2	<i>1 mark for each point.</i> • Extracting less energy as electricity leaves the cooling / condenser water at a higher temperature (1) • Hotter water is more useful for district heating of homes, fish farms, greenhouses (1)	2
03.3	<i>1 mark.</i> • Hot water loses heat and needs pumping energy over distance; heat cannot be transported as far as electricity (1)	1
03.4	<i>1 mark for each point.</i> • Long narrow pipes give a large surface area and copper is a good thermal conductor so heat transfers quickly (1) • Hot and cold fluids flowing in opposite directions maintain a temperature gradient along the whole length so more heat is transferred (1) • Slower flow allows more time for heat to transfer (1)	3

Question	Marking guidance	Marks
03.5	<i>1 mark for each point.</i> - One large tank has a lower surface area : volume ratio than many small ones (1) - A sphere has the smallest surface area for any volume; less surface means less heat loss (with insulation of low thermal conductivity) (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 233-235; AQA spec 3.3.4.2.3.	
04.1	<i>1 mark for the correct answer.</i> 25 g ha⁻¹ (accept 24-27) (1)	1
04.2	<i>1 mark.</i> The highest dose that can be applied without killing any members of the population / any individuals of the (non-target) species (1)	1
04.3	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> - The MDAF (minimum dose always fatal) of the pest is about 50 g ha⁻¹ (1) - This is higher than the MDNF of the bee (about 20 g ha⁻¹) (1) - So a dose that kills all pests kills some bees / the two curves overlap between about 20 and 50 g ha⁻¹ (1) - A dose of 30-40 g ha⁻¹ would kill most pests with limited bee deaths (compromise) (1)	3
04.4	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> - Spray at night / when bees are not flying (1) - Spray when the crop is not in flower (1) - Spray on still days to reduce spray drift (1) - Use a systemic pesticide / seed coating rather than spraying (1) - Use smaller droplets for more even coverage (1)	2
04.5	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> - Specificity is the extent to which the pesticide is toxic only to the target species / range of species affected (1) - A highly specific pesticide has low toxicity to non-target species (eg bees, fish, mammals) (1) - So fewer non-target deaths / less loss of predators, pollinators (1) - Fewer indirect (food chain) effects / fewer inter-species relationships lost (1) - Example: pyrethroids have high insect toxicity but low mammal toxicity (1)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 262-263; AQA spec 3.4.3.2.8.	
05.1	<i>A correct final answer (with unit where asked) scores full marks. If the answer is wrong, award 1 mark for each correct step of working shown.</i> - Water = 65.0 - 55.0 = 10.0 g; fresh soil = 40.0 g (1) 10.0 / 40.0 x 100 = 25.0% (1)	2
05.2	<i>A correct final answer (with unit where asked) scores full marks. If the answer is wrong, award 1 mark for each correct step of working shown.</i> - Organic matter = 55.0 - 52.6 = 2.4 g; dry soil = 30.0 g (1) 2.4 / 30.0 x 100 = 8.0% (1)	2
05.3	<i>1 mark for each point.</i> - So all water is removed and the mass loss on burning is due to organic matter only (1) - Heating until mass no longer changes shows all water has evaporated (1)	2

Question	Marking guidance	Marks
05.4	<i>1 mark for each point.</i> - Organic matter releases nutrients as it decomposes and provides food for soil biota (1) - Increases water retention and, as humus, binds particles into crumb peds improving structure / aeration (1) - Surface organic matter / humus adhesion reduce rain splash and wind erosion (1)	3
05.5	<i>1 mark.</i> Samples taken from different depths / locations (organic matter is concentrated near the surface); incomplete burning; different heating times (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 158-159, 412; AQA spec 3.2.5.1, 3.7.2.1.	
06.1	<i>1 mark for each point.</i> - Quadrat size depends on the size / spacing of the organisms: 0.5 m x 0.5 m is typical for ground flora (1) 10 cm x 10 cm is for lichens (too small to contain a representative sample of plants); 100 m x 100 m is for trees (impractical, too much work) (1)	2
06.2	<i>A correct final answer (with unit where asked) scores full marks. If the answer is wrong, award 1 mark for each correct step of working shown.</i> $N = 240$; $N(N-1) = 240 \times 239 = 57\,360$ (1) $\Sigma n(n-1) = 48 \times 47 + 120 \times 119 + 36 \times 35 + 24 \times 23 + 12 \times 11 = 2256 + 14280 + 1260 + 552 + 132 = 18\,480$ (1) $D = 57\,360 / 18\,480 = 3.10$ (accept 3.1) (1)	3
06.3	<i>1 mark for each point.</i> - Lower D means lower biodiversity: the ungrazed meadow is dominated by one or a few species (eg tall grass) with the others rare (less even) (1) - Grazing (plagioclimax management) prevents dominance by vigorous species so abundance is more evenly spread (1)	2
06.4	<i>1 mark for each point.</i> - A frame quadrat flattens tall plants, making percentage cover estimates inaccurate and hiding shorter plants (1) - Point quadrat pins are lowered through the vegetation; each touch counts (10 pins = 10% each) without flattening (1)	2
06.5	<i>1 mark.</i> Subjective judgement varies between recorders; results are not quantitative so statistical analysis is not possible (but quick for many areas) (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 394-395, 401-402; AQA spec 3.7.2.1.	

Question	Marking guidance	Marks
07.1	<i>1 mark for each point.</i> • Double hull: a gap of up to 2 m between the hulls so damage to the outer hull in a collision / grounding does not release oil (1) • Inert gas system: cooled engine exhaust gas (no oxygen) fills the space above the oil so an explosive oil vapour / air mixture cannot form (1) • Separate ballast tanks: ballast water is not carried in oil tanks so no residual oil is discharged when ballast is pumped out (1) • Twin engines / rudders: if one fails the ship can still be steered / propelled and is not carried onto rocks by currents or wind (1) • Crude oil washing: sludges are mixed and removed with the cargo so tanks do not need washing with seawater that was then discharged; oily water is unloaded at the terminal (1)	5
07.2	<i>1 mark for each point.</i> • Offshore routes keep tankers further from the coast so if there is engine failure there is more time to deal with it before the ship is carried ashore (1) • GPS gives accurate navigation; AIS lets ship positions be monitored and coordinated to avoid collisions and groundings (1)	2
07.3	<i>1 mark for each point.</i> • The tank stands on an impermeable base surrounded by a wall (1) • The volume enclosed by the wall is greater than the tank volume so all the oil is contained if the tank splits / cannot overflow (1)	2
07.4	<i>1 mark.</i> • Separates floating oil from drainage water so the oil is retained (for later treatment) and only water flows to the drains / river (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 259-260; AQA spec 3.4.3.2.7.	
08.1	<i>Correct answer only.</i> • B - Tropospheric ozone (formed by reactions of NO_x in sunlight) (1)	1
08.2	<i>1 mark for each point.</i> • Primary pollutant: released directly by human activities, eg sulfur dioxide from burning coal / smelting sulfide ores; NO_x from high-temperature combustion (1) • Secondary pollutant: produced by chemical reactions between primary pollutants, often with non-pollutants (water, oxygen, sunlight) (1) • Sulfur dioxide is oxidised by ozone to sulfur trioxide which dissolves to form sulfuric acid; NO_x dissolve to form nitric acid (1) • The acids (sulfurous, sulfuric, nitric) are secondary pollutants; ozone itself is a secondary pollutant of NO_x photochemistry (1)	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 237-238; AQA spec 3.4.1.	
09.1	<i>1 mark for each point.</i> • A fluid in a buried ground loop absorbs low-temperature heat from the ground; at the expansion valve the fluid boils at low pressure absorbing this heat (1) • A compressor raises the pressure so the gas condenses, releasing heat at a higher temperature inside the building (1) • The electricity only moves heat rather than creating it, so up to four times as much heat is delivered as electricity used (less if the source is colder) (1)	3

Question	Marking guidance	Marks
09.2	<i>1 mark for each point.</i> • Parabolic reflectors focus sunlight onto tubes of oil, increasing the energy density and reaching high temperatures (1) • The heat is transferred to molten salt (potassium nitrate) stored in large insulated tanks at up to 550 °C (1) • The stored heat boils water to drive steam turbines whenever electricity is required, including at night (1)	3
09.3	<i>1 mark for each point.</i> • Reflectors can only focus parallel rays of direct sunlight; cloud, dust and smoke scatter light so it cannot be concentrated (1) • PV cells convert diffuse as well as direct light, though output is lower (1)	2
09.4	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Manufacture: extraction / processing of silicon, metals and plastics; toxic wastes (silicon tetrachloride, cadmium) (1) • Land take by large solar farms competing with other uses / affecting desert flora and fauna (1) • Water for cleaning panels in arid areas (1) • Disposal of old panels (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 200-202; AQA spec 3.3.4.1.	

Question	Marking guidance	Marks
10.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Control of greenhouse gases: energy conservation (transport, buildings, industry), renewable and nuclear energy replacing fossil fuels, CCS, reduced deforestation and afforestation (carbon sequestration), methane capture from landfill / digesters, HFC controls, changes in agriculture • Geoengineering: white roofs, ocean fertilisation, solar shades - largely untried, possible unpredictable consequences • International agreements: Kyoto Protocol, Paris Agreement - targets, but non-binding / uneven participation; carbon trading and taxes; comparison of per capita carbon footprints between countries • Evidence of limited success: global CO₂ concentrations still rising (about 280 ppm pre-industrial to over 410 ppm); temperatures up about 1 °C; emissions in some developed countries falling but rising in developing economies • Reasons for difficulty: fossil fuels underpin economies, need for major lifestyle change, cost, time delays and natural fluctuations masking effects, political disagreement, developing countries' right to develop • Adaptation strategies (flood defences, drainage, managed retreat, infrastructure changes) reduce impacts but do not address the cause • Comparison with the Montreal Protocol's success for ozone • Judgement: partial success - technology exists and some regional reductions achieved, but global strategies have not yet stopped the rise; tipping points make urgency greater	9
	Model answer - Level 3 (8 marks)	
	Strategies fall into control of greenhouse gases, geoengineering, international agreements and adaptation. Emissions are cut by energy conservation in transport, buildings and industry, by replacing fossil fuels with renewables and nuclear, by carbon capture and storage, by reducing deforestation and planting forests, and by capturing methane from landfill. Geoengineering ideas - white roofs, ocean fertilisation, solar shades - remain largely untried with unpredictable side effects. Adaptation, such as flood defences and managed retreat, reduces impacts but not the cause.	

Question	Marking guidance	Marks
	The Kyoto Protocol and Paris Agreement set targets, but they are weak or non-binding and participation is uneven; carbon trading and taxes exist but per capita footprints still differ enormously between countries. The evidence of success is limited: atmospheric CO ₂ has risen from about 280 ppm before industrialisation to over 410 ppm and temperature by about 1 °C, and although emissions are falling in some developed countries they are rising in developing economies.	
	The difficulty is that fossil fuels underpin economies, so control demands lifestyle change, cost, and political agreement between nations with different priorities, while time delays and natural fluctuations mask the effect of any action. By contrast the Montreal Protocol succeeded because ozone was a substitution problem. The judgement must be partial success: the technology exists and regional reductions prove it works, but global strategies have not yet stopped the rise, and tipping points make the delay dangerous.	
	Weaker answer - Level 2 (5 marks)	
	Methods to reduce climate change include using renewable energy such as wind and solar instead of fossil fuels, saving energy, and planting trees to absorb carbon dioxide. International agreements like the Kyoto Protocol and the Paris Agreement have set targets for countries to reduce emissions. However carbon dioxide levels are still rising, so the methods have not been very successful so far. This is because countries rely on fossil fuels and do not always keep to the agreements. More needs to be done.	
	What makes the difference	
	• Level 2 covers the main strategies and reaches a conclusion, but with no quantitative evidence (280 to 410 ppm, 1 °C) and no mention of geoengineering or adaptation.	
	• Level 3 explains <i>why</i> success is limited (economic dependence, time lags, natural variability, political disagreement) rather than just stating that it is.	
	• The Montreal comparison and the tipping-point point give Level 3 the breadth and judgement the top band requires.	
10.2	<i>1 mark.</i> • The total greenhouse gas / CO₂ emissions caused by an individual, organisation, product or country (often expressed per capita per year) (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 89-107; AQA spec 3.2.1.2.	

Question	Marking guidance	Marks
11.1	Discuss how knowledge of the properties of energy resources can be used to reduce the severity of their impact on the environment. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Properties: energy density, renewability, abundance, location, intermittency / predictability, storage, transport, conversions, technology level, environmental impacts • Fossil fuels: choosing lower-impact fuels (gas over coal), desulfurisation, CCS, efficient combustion, siting power stations near cooling water and coal ports; new extraction technologies (directional drilling, secondary / tertiary recovery) and their impacts (fracking, tar sands) • Nuclear: high energy density reduces transport; waste properties (half-life) determine storage; thorium and fusion prospects • Renewables: matching resource to location (wind, HEP, solar, wave, tidal, geothermal, biofuel siting); intermittency managed by storage (pumped HEP, batteries, hydrogen, P2G, V2G), grid management and diversity; low-impact designs (helical turbines, VAWT, tidal lagoons) • Energy conservation reduces demand: transport, buildings, industry (CHP, heat exchangers) • Impacts before and during use; life-cycle / embodied energy • Evaluation: no resource is impact-free; knowledge allows the best mix and mitigation; judgement Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 175-235; AQA spec 3.3.2-3.3.4.	25

Question	Marking guidance	Marks
11.2	Discuss the strategies that can be used to provide sustainable supplies of water for a growing population. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Context: uneven distribution of water, growing population and per capita demand, unsustainable exploitation of rivers and aquifers, hydrological cycle links • New sources: rainwater collection, rivers (discharge, fluctuations, contaminants), reservoirs and site selection, estuary barrages, desalination, unexploited aquifers • Management: artificial aquifer recharge, river-regulation reservoirs, inter-basin transfers, afforestation • Conservation: low-water appliances, xeriscaping, drip irrigation, grey water, leak reduction, metering; pollution control to protect sources • Treatment: matching quality to use; screens, sedimentation, aeration, flocculation, filtration, sterilisation, pH adjustment, activated carbon, reverse osmosis • Evaluation of costs, energy, environmental impacts (habitat loss, methane, inter-tidal habitats, conflicts) and justified conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 114-127; AQA spec 3.2.2.4-3.2.2.5.	25